

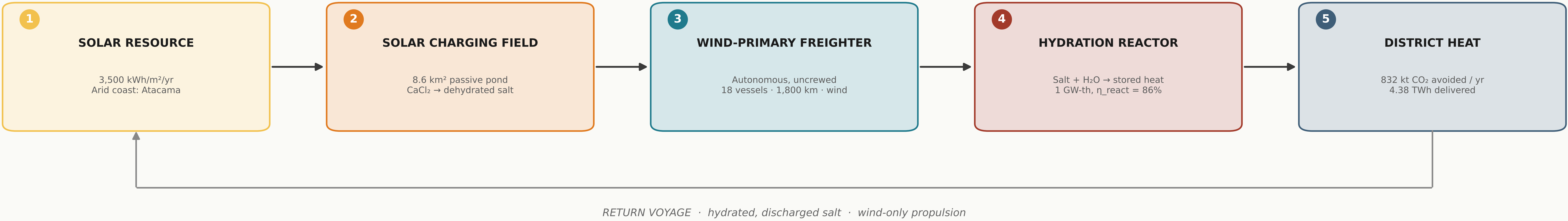
Solar Thermochemical Heat Export via Closed-Loop Salt Hydration

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The discharge temperature is the use temperature — no electricity round-trip, no exergy loss.

Dotted Labs · Concept paper v0.3 · 2026

Reference loop · CaCl₂ passive pond · Atacama → Southern Cone, 1,800 km



\$54 / MWh

Delivered heat (LCOH) on the Atacama → Southern Cone reference route, inside the \$40–66 target band — no carbon price assumed.

80 %

Round-trip thermal efficiency, sun-on-pond to delivered-heat. No electricity round-trip; the discharge temperature is the use temperature.

1.6 Gt CO₂

Per-year carbon wedge achievable by 2050 in the Fast deployment scenario — ≈ 80 % of the IEA NZE low-temperature heat decarbonisation gap.

WHAT IS NEW

Closed-loop maritime thermochemical heat export has not, to our knowledge, been modeled end-to-end. Adjacent literature treats either (i) stationary salt storage at the plant, or (ii) hydrogen / ammonia carriers.

Salt-hydrate carriers are uniquely suited to low-temperature district heat: the discharge temperature is the use temperature. There is no exergy-destroying conversion to electricity or chemical fuel — sunlight → bonds → heat, full stop.

The v0.3 architectural pivot drops the v0.2 cargo-as-fuel auxiliary engine in favour of autonomous wind-primary freighters. Long-haul LCOH roughly halves; the highest-carbon-leverage routes (Pilbara → N. China, Namibia → NW Europe) become economically tractable.

CANDIDATE WORKING FLUIDS

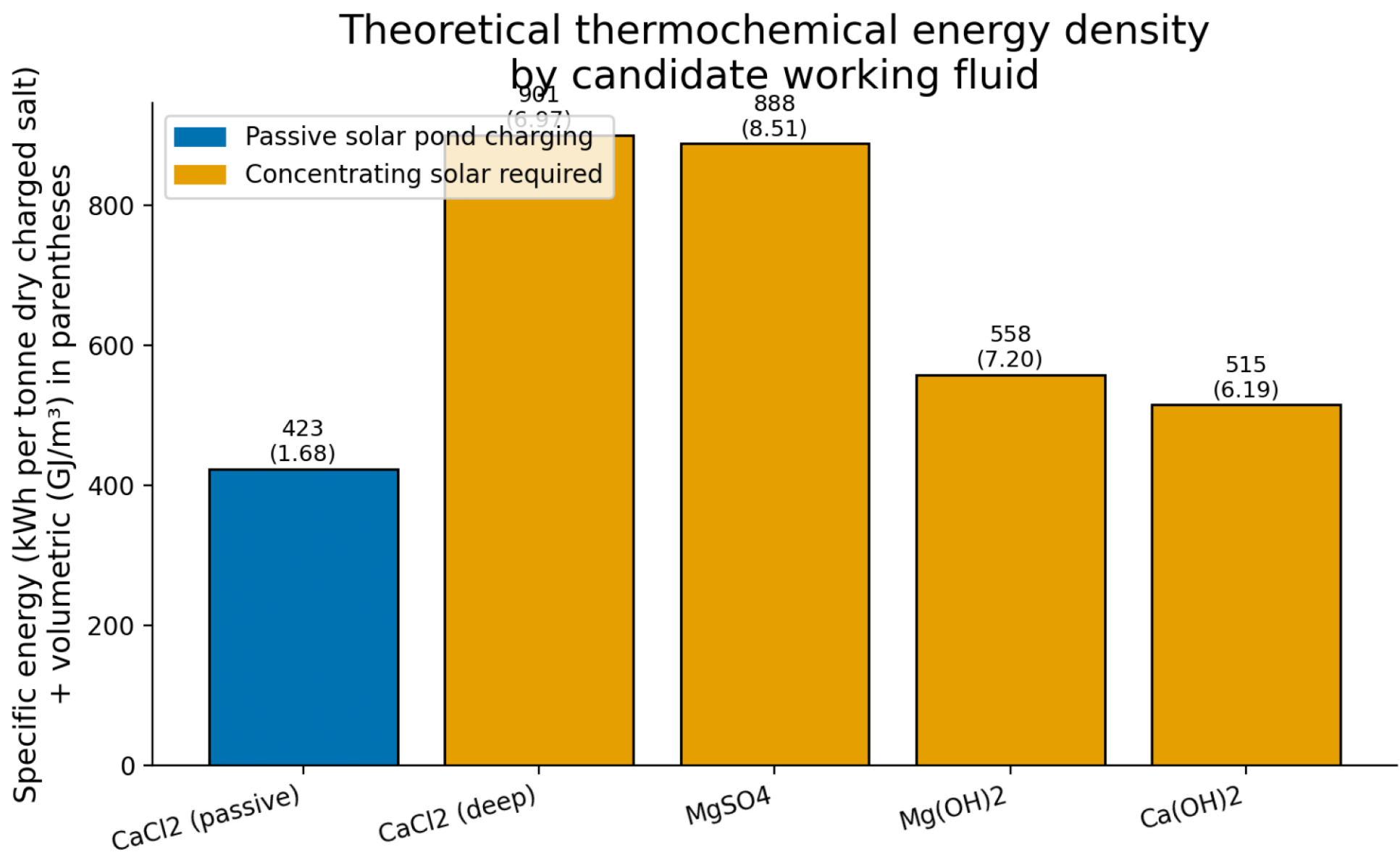


Fig. 1 — Theoretical energy density of candidate salt-hydrate pairs.

CHEMISTRY × ROUTE GRID

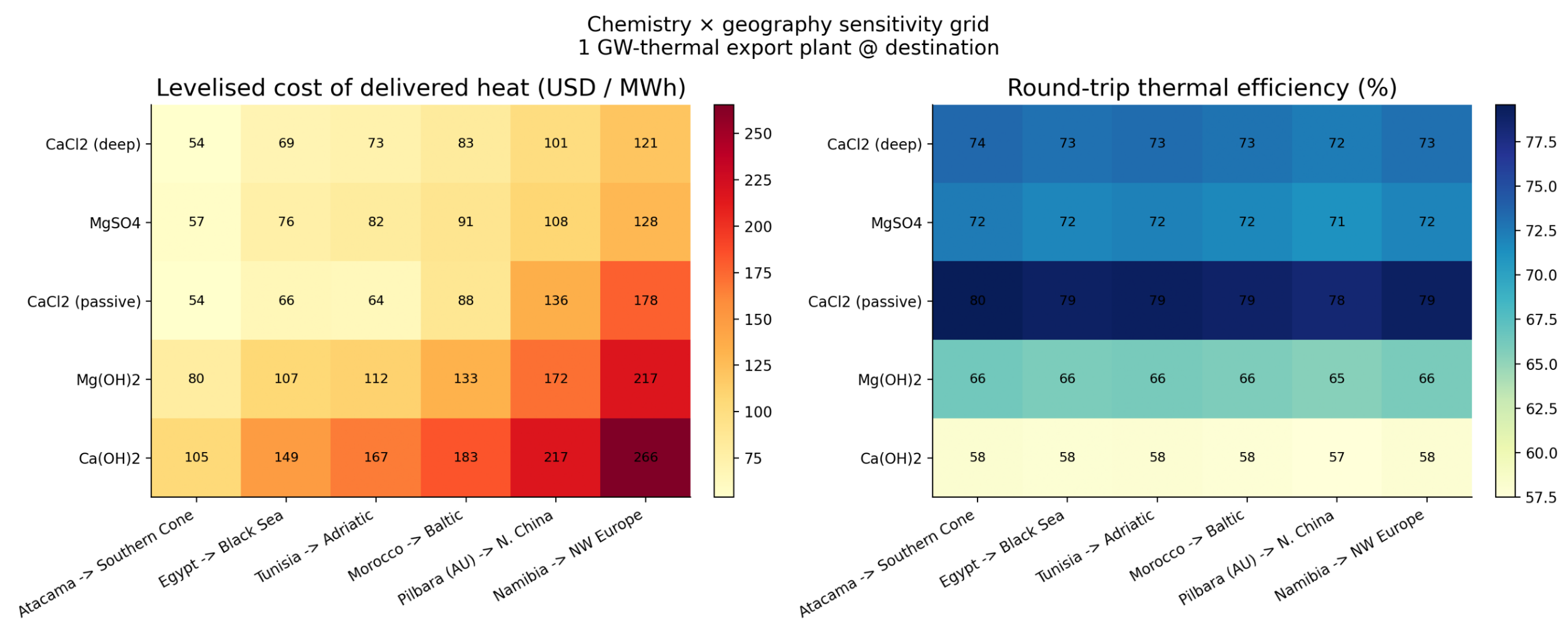


Fig. 2 — LCOH (left) and round-trip thermal efficiency (right) across the chemistry × route matrix.

LCOH VS VESSEL CRUISE SPEED

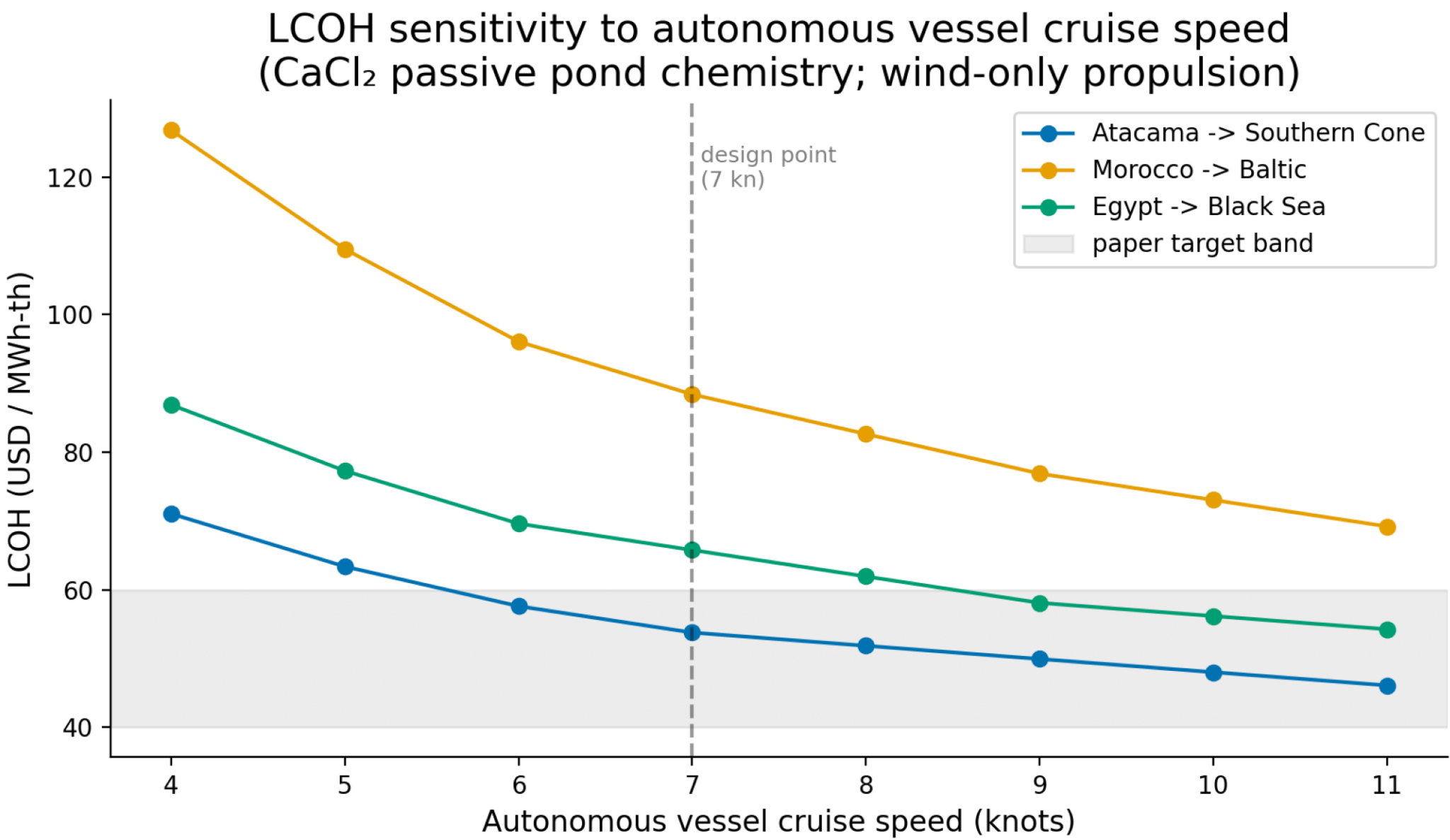


Fig. 4 — LCOH sensitivity to autonomous vessel cruise speed (≈ \$5 / MWh per knot, across the route set).

CLIMATE LEVERAGE AT SCALE

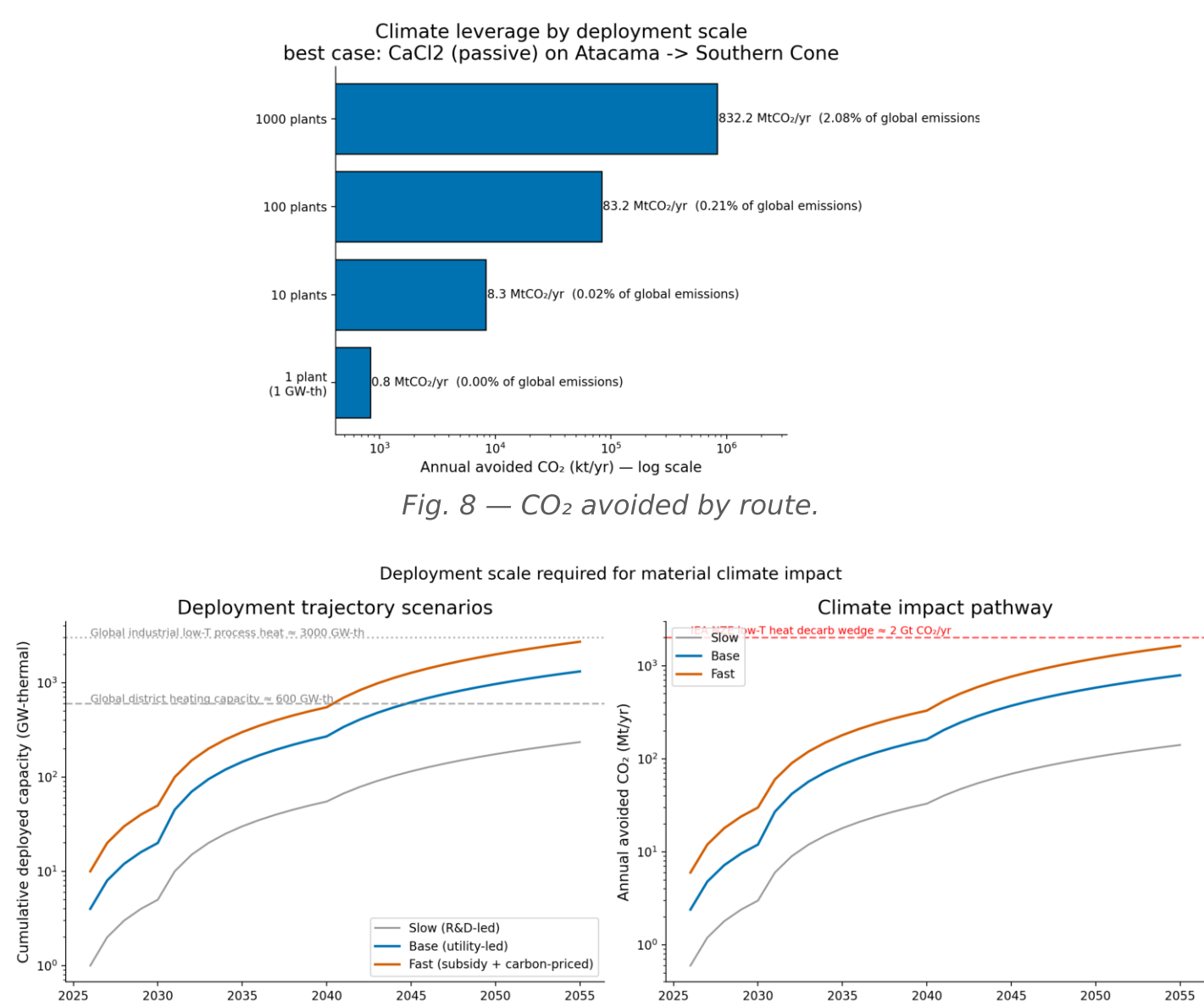


Fig. 8 — CO₂ avoided by route.

Fig. 11 — Fast scenario reaches ≈ 80 % of IEA NZE low-T heat wedge by 2050.

FINDINGS · OPEN QUESTIONS · NEXT STEPS

Findings

- Two viability tiers, not one. Short-haul passive-pond CaCl₂ clears \$54–66/MWh; long-haul deep-CSP CaCl₂ sits at \$100–120/MWh.
- Cargo-as-fuel was the v0.2 showstopper. Autonomous wind-primary shipping eliminates it; round-trip efficiency rises to a uniform 58–80%.
- Freshwater coproduction (2–6 Mm³/yr per GW-th plant) is a real, monetisable co-revenue stream in arid source regions.

Binding open questions

- Multi-cycle salt stability (≥ 1,000 cycles). Bench item.
- Autonomous wind-cargo fleet scale-up (5,000+ vessels by 2050).
- Site-specific wind statistics in the cruise-speed model.

Next steps

- 200-cycle CaCl₂ bench validation at a partner lab.
- Vendor engagement (Ladon Robotics or peer) for vessel-class specs.
- 100 kW-th pilot at Mejillones, Chile.

Selected references

N'Tsoukpoe et al. 2009 · Donkers et al. 2017 · Trausel et al. 2014 · Cot-Gores et al. 2012 · IEA NZE 2023.